

S.T.A.R. Experiment Registry

Version 0.2

Operational Specification for Arithmetic–Geometric–Topological Investigation of Cosmic Structure

1. Purpose

The S.T.A.R. Experiment Registry defines the controlled experimental program required to evaluate the hypotheses, mathematical constructions, phenomenological mappings, and theoretical extensions identified in the S.T.A.R. Research Charter and Claim/Evidence Registry.

The registry converts research claims into reproducible experiments with explicit:

- experiment identifiers;
- associated claims;
- scientific questions;
- datasets;
- data provenance requirements;
- independent and dependent variables;
- mathematical constructions;
- mapping choices;
- locked parameters;
- null models;
- train/test or calibration/validation design;
- statistical tests;
- success and failure conditions;
- reproducibility requirements;
- historical evidence status;
- reconstruction requirements;
- replication requirements.

The registry is intended to prevent the project from moving directly from a mathematical construction to an interpretation of physical reality without an intermediate experimental chain.

The governing principle is:

Do not ask whether the theory can be made to fit the universe. Ask whether a fixed mathematical construction predicts structure that was not used to construct it.

2. Relationship to the Research Charter

The Experiment Registry implements the experimental portion of the canonical S.T.A.R. pipeline:

```
[  
E  
  \rightarrow  
  I(E)  
    \rightarrow  
    \Phi(E)  
      \rightarrow  
      \mathcal{H}_{\rm arith}  
        \rightarrow  
        D_{\rm arith}  
          \rightarrow  
          D_{\rm cosmic}  
            \rightarrow  
            \text{Statistical Test}  
              \rightarrow  
              \text{Independent Validation}.  
]
```

where:

- (E) = arithmetic object or mathematical source;
- $I(E)$ = arithmetic invariants;
- $(\Phi(E))$ = arithmetic-to-geometric mapping;
- $(\mathcal{H}_{\rm arith})$ = arithmetic/geometric construction;
- $(D_{\rm arith})$ = arithmetic-derived data;
- $(D_{\rm cosmic})$ = astronomical observations;
- statistical testing evaluates correspondence;
- independent validation evaluates whether the correspondence survives outside the construction dataset.

The registry therefore sits between the Claim/Evidence Registry and the Results/Replication Registries.

3. Epistemic Status

Every experiment must distinguish the status of the claim being tested.

D — Derived

Mathematically derived from stated assumptions.

E — Empirical

Observed in an astronomical or simulated dataset.

M — Model

A phenomenological or computational construction whose physical interpretation remains open.

P — Postulate

An explicitly assumed component of the framework.

H — Hypothesis

A proposition being tested.

C — Conjecture

A broader mathematical or physical proposition not yet established.

T — Testable Prediction

A quantitative prediction whose outcome can discriminate between hypotheses or null models.

S — Speculative

A possible interpretation or future extension that is not currently experimentally established.

An experiment may test multiple epistemic levels, but the registry must not collapse them into a single status.

4. Evidence-State Classification

Each experiment receives an evidence-state designation.

U — Usable Historical R&D

Existing R&D contains sufficiently specific information to support direct use or limited reconstruction.

R — Reconstruction Required

Historical work exists, but preprocessing, leakage control, parameter locking, provenance, or statistical design must be reconstructed before the result can be treated as controlled evidence.

N — New Experiment Required

Existing R&D does not constitute a controlled experiment sufficient for the claim.

C — Control

The experiment exists primarily to establish a baseline, null, or alternative explanation.

D — Derivation Audit

The primary task is mathematical derivation or consistency checking.

H — Historical Only

The material is preserved for research lineage but is not treated as current evidence.

X — Negative/Diagnostic

The result is useful because it identifies instability, failure, sensitivity, or a limitation.

5. Standard Experiment Record

Every experiment in this registry should eventually contain the following fields.

Field

Requirement

Experiment ID	Unique permanent identifier
Experiment Name	Descriptive title
Priority	A / B / C
Associated Claim IDs	Claim/Evidence Registry references
Research Question	Question being tested
Hypothesis	Explicit testable proposition
Null Hypothesis	Default competing explanation
Dataset ID	Controlled data reference
Data Provenance	Source and processing lineage
Population	Objects included
Exclusions	Pre-registered exclusions
Independent Variables	Predictor/construction variables
Dependent Variables	Outcomes
Arithmetic Inputs	Invariants used
Mapping	(\Phi) specification
Parameters	Locked values/ranges
Controls	Baseline and null controls
Train/Test Design	If applicable
Statistical Test	Primary test
Secondary Tests	Additional diagnostics
Success Criterion	Predefined
Failure Criterion	Predefined
Reproducibility	Required artifacts
Historical R&D	Existing supporting work

Reconstruction Status	None / Partial / Complete
Replication Dataset	Independent validation
Current Status	Planned / Running / Complete / Failed / Superseded

6. Priority Structure

Priority A — Core Empirical Program

Experiments directly addressing the primary research questions:

1. arithmetic/rank and cosmic environment;
2. arithmetic/rank and cosmic topology;
3. arithmetic features and SFR;
4. mapping-family discrimination;
5. null/permutation controls;
6. independent astronomical replication.

Priority B — Theoretical/Physical Extension

Experiments concerning:

- RTCH;
- arithmetic fields;
- entropy/cohomology;
- generalized mass-energy relations;
- dynamical/evolutionary extensions;
- physical constraints.

Priority C — Exploratory Extensions

Experiments that may become relevant after the core program is validated, including:

- alternative arithmetic invariant families;
- alternative symbolic mappings;
- additional astrophysical observables;
- quantum-level constructions;

- Hubble–Planck tension applications.
-

7. EXP-CTRL-A03 — Leakage and Contamination Audit

Priority: A

Evidence State: R

Associated Claims: CORE-001, SFR-A001, COSMO-L001

Objective

Determine whether historical S.T.A.R. machine-learning and feature-engineering experiments contain information leakage between training and testing populations.

Historical motivation

Several historical scripts construct:

- normalized variables;
- redshift bins;
- $(L_{\rm cosmo}(s))$;
- `cosmo_rank`;
- interaction terms;

before the train/test split.

Therefore historical performance values cannot automatically be interpreted as leakage-free predictive measurements.

Procedure

Reconstruct the relevant pipelines so that:

```
[
D_{\rm train}
\rightarrow
\text{fit preprocessing}
\rightarrow
```

$D_{\text{train}}^{\prime}$,
]

and the fitted transformations are subsequently applied to:

[
 D_{test}
 $\rightarrow$
 $D_{\text{test}}^{\prime}$.
]

No target-derived or population-derived quantity may be calculated using the complete dataset before partitioning.

Required controls

Compare:

1. historical pipeline;
2. leakage-controlled pipeline;
3. shuffled-target pipeline.

Success Criterion

The reconstructed pipeline must reproduce the historical result within a documented tolerance where no leakage is present.

Failure Criterion

Any material performance degradation attributable to information leakage must be recorded and the historical result reclassified accordingly.

8. EXP-CTRL-A01 — Random-Feature Baseline

Priority: A

Evidence State: C/R

Associated Claim: CTRL-RF001

Objective

Establish a non-theoretical machine-learning baseline.

Historical basis

`model15.py` used random feature selection and a fixed random seed for predictive modeling.

Design

Construct a baseline using randomly selected astrophysical features under a fixed seed.

Compare against:

- ordinary astrophysical baseline;
- arithmetic-feature baseline;
- arithmetic-plus-astrophysical model.

Primary endpoint

Out-of-sample:

[
 $R^2_{\rm OOS}$.
]

Secondary:

[
 $\{\rm RMSE\}, \quad \{\rm MAE\}$.
]

Purpose

Determine whether an apparent improvement from S.T.A.R. features exceeds ordinary feature-selection effects.

9. EXP-CTRL-A02 — Arithmetic Permutation Control

Priority: A

Evidence State: N

Associated Claims: SFR-A001, ARITH-001

Objective

Determine whether arithmetic-derived features contain information beyond their marginal distributions and accidental correlations.

Procedure

Preserve the marginal distribution of each arithmetic feature while breaking its association with the astronomical target.

Possible permutation procedures include:

$\pi(X_{\text{arith}})$.
]

Where appropriate, permutations should preserve relevant grouping variables.

Comparison

[
 X_{astro}
]

versus

[
 $X_{\text{astro}} + X_{\text{arith}}$
]

versus

[
 $X_{\text{astro}} + X_{\text{arith}}^{\text{perm}}$.
]

Primary endpoint

[
 $\Delta R^2_{\text{rm OOS}}$.
]

Success Criterion

The unpermuted arithmetic model must outperform the appropriately constructed permutation distribution at a pre-registered significance level.

10. EXP-K001 — Log-Normal Reciprocal-Normalization Test

Priority: A

Evidence State: R

Associated Claim: MATH-K001

Historical construction

$$\sum_{i=1}^N \left(\frac{M_0}{M_i} \right)^s$$

with normalization

$$\frac{\sum_i (M_i/M_0)}{\sum_i (M_0/M_i)}.$$

For a log-normal mass distribution with

[
 $\log M_i \sim N(\mu, \sigma^2),$
]

and

$$[M_0=e^{\mu},]$$

the historical derivation gives equal limiting expectations for the numerator and denominator, suggesting

$$[K_{\rm theory}\rightarrow 1.]$$

Objective

Separate this statistical result from its historical BSD-inspired interpretation.

Experimental groups

1. Gaussian-log mass distribution;
2. skewed distribution;
3. heavy-tailed distribution;
4. mixture distribution;
5. empirical galaxy-mass distributions.

Variables

$$[N,\quad \mu,\quad \sigma,\quad K.]$$

Primary analysis

Estimate:

$$[E[K_N],\quad \{\rm Var\}(K_N)]$$

as a function of (N).

Historical observations

The R&D archive contains simulations with approximately:

[
N=500,\quad
\mu=9.7843,\quad
\sigma=0.3645,
]

and reported mean:

[
K\approx0.998108.
]

Historical observational samples also produced materially different values, including small-(N) results.

These are treated as reconstruction targets, not established confirmations.

Success Criterion

The theoretical convergence behavior is reproduced under the stated assumptions and its finite-sample behavior is characterized.

11. EXP-L001 — ($L_{\rm cosmo}(s)$) Model Validation

Priority: A

Evidence State: R

Associated Claim: COSMO-L001

Hypothesis

($L_{\rm cosmo}(s)$) may contain information about galaxy-population structure beyond ordinary mass/redshift variables.

Core construction

$$\sum_i \left(\frac{M_0}{M_i} \right)^s.$$

Historical implementations also introduced redshift binning and derived mass-dependent quantities.

Critical requirement

All population statistics and normalization parameters must be estimated within the training fold.

Baselines

1. mass-only;
2. mass + redshift;
3. standard astrophysical feature set;
4. astrophysical + ($L_{\rm cosmo}$);
5. shuffled ($L_{\rm cosmo}$).

Primary endpoint

Incremental out-of-sample predictive information:

$$[\Delta R^2_{\rm OOS}].$$

Secondary endpoints

$$[\Delta \{\rm RMSE\}, \quad \Delta \{\rm MAE\}, \quad \text{calibration error}].$$

Success

($L_{\rm cosmo}$) demonstrates reproducible incremental information on independent data.

12. EXP-SFR-A01 — Arithmetic Incremental SFR Prediction

Priority: A

Evidence State: N/R

Associated Claim: SFR-A001

Hypothesis

Arithmetic-derived variables may provide information about SFR beyond ordinary astrophysical variables.

Target

[
 $Y = \log\{\mathrm{SFR}\}_{\mathrm{H}\alpha}.$
]

Baseline

[
 $Y = f(X_{\mathrm{astro}}).$
]

Extended model

[
 $Y = f(X_{\mathrm{astro}}, X_{\mathrm{arith}}).$
]

where (X_{arith}) may include:

[
 $r_E,$
 $\log|\Delta_E|,$
 $\log N_E,$
 $\log\mathrm{operatorname{Reg}}(E),$
 $T_E,$
]

and validated mapping-derived variables.

Primary endpoint

$R^2_{\text{arithmetic+astro}}$

R^2_{astro} .

]

Secondary endpoints

- RMSE;
- MAE;
- residual distributions;
- calibration;
- SHAP stability;
- performance across SFR strata.

Critical control

Arithmetic permutation control EXP-CTRL-A02.

Success Criterion

An incremental effect survives:

- leakage-controlled preprocessing;
- held-out testing;
- permutation control;
- multiple random seeds;
- independent dataset validation.

A statistically significant effect without practical predictive value is not sufficient for Level 4 predictive correspondence.

13. EXP-SFR-A02 — Arithmetic-Only SFR Prediction

Priority: A
Evidence State: N
Associated Claim: SFR-A002

Objective

Test whether arithmetic variables alone contain predictive information about SFR.

Model

[
 $Y=f(X_{\rm arith})$.
]

Compare against:

[
 $Y=f(X_{\rm astro})$
]

and shuffled arithmetic controls.

Purpose

This experiment distinguishes:

1. genuine arithmetic information;
2. arithmetic features merely acting as proxies for known astrophysical variables;
3. random correlations.

14. EXP-SFR-A03 — Extreme-SFR

Robustness

Priority: A
Evidence State: R/N
Associated Claim: SFR-E001

Historical motivation

Historical models showed substantially different behavior for extreme-SFR populations compared with the bulk population.

One historical analysis reported approximately:

- full-sample RF ($R^2 \approx 0.8455$);
- full-sample GB ($R^2 \approx 0.8476$);
- non-outlier RF ($R^2 \approx 0.8729$);
- non-outlier GB ($R^2 \approx 0.8766$);

with substantially poorer performance in the extreme/outlier population.

These historical values require reconstruction under controlled preprocessing.

Design

Pre-register SFR strata rather than defining outliers after observing model performance.

Evaluate:

[
 R^2 , RMSE , MAE
]

within each stratum.

Objective

Determine whether arithmetic features retain any incremental information specifically in extreme-SFR populations.

15. EXP-RANK-A01 — Rank and Cosmic Environment

Priority: A

Evidence State: N

Associated Claims: RANK-001, RANK-002

Hypothesis

Elliptic-curve rank may exhibit statistically nontrivial correspondence with cosmic environment.

Test variables

Rank:

```
[  
r_E.  
]
```

Environmental variables may include:

- galaxy density;
- local density;
- cluster membership;
- filament membership;
- void membership;
- environmental classification.

Primary analysis

Estimate:

```
[  
P(r_E\mid{\rm environment})  
]
```

and compare against appropriate null distributions.

Nulls

- random assignment of elliptic curves to galaxies;
- permutation of rank labels;
- matched environmental sampling;
- alternative arithmetic invariants.

Important distinction

The experiment does not assume:

```
[  
r=0\Rightarrow{\rm void},  
]
```

or any other one-to-one physical interpretation.

Those are hypotheses to be tested.

16. EXP-RANK-A02 — Rank-Weighted Cosmic Topology

Priority: A

Evidence State: N/R

Associated Claims: RANK-003, TOPO-001, TOPO-002

Hypothesis

Arithmetic rank may contribute information about cosmic-web topology.

Construction

Define a rank-weighted filtration:

[
 $K_{\epsilon^{\{r\}}}$.
]

The exact construction must be fixed before the final comparison.

Topological observables

Persistent homology:

[
 $\beta_0, \beta_1, \beta_2$.
]

Interpretation:

- β_0 : connected-component structure;
- β_1 : one-dimensional cycles/connectivity;
- β_2 : cavities/void structure.

Persistent homology must not be interpreted as a direct filament counter.

Filaments

Filament identification should primarily use:

- geometric skeleton methods;
- graph methods;
- established cosmic-web classifiers.

Persistent homology provides supporting topological information.

Comparison

[
D_{\rm arith}
\leftrightharpoonrightarrow
D_{\rm cosmic}.
]

Use:

- Wasserstein distance;
- bottleneck distance;
- persistence landscapes/images;
- topology-aware statistics.

Null models

- randomized rank assignments;
- randomized arithmetic mappings;
- spatial shuffling;
- density-preserving controls.

Historical values such as:

[
W_2<10^{-2}
]

are exploratory/model-dependent and are not accepted as universal evidence thresholds.

17. EXP-TOPO-A01 — Null-Calibrated Persistent Homology

Priority: A

Evidence State: R/N

Associated Claims: TOPO-001, TOPO-002, TOPO-003

Objective

Establish whether any observed arithmetic/topological correspondence exceeds what can be produced by the null models.

Required analysis

For each mapping:

[
D_{\rm observed}
]

must be compared against an empirical null distribution:

[
D_{\rm null}^{\{1\}}, \ldots, D_{\rm null}^{\{N\}}.
]

Output

Report:

- effect size;
- empirical (p)-value;
- confidence interval;
- null distribution;
- multiple-comparison correction where applicable;
- sensitivity to resolution;
- sensitivity to sampling density;
- sensitivity to noise;
- mapping sensitivity.

Success Criterion

Observed correspondence must survive the pre-registered null ensemble.

18. EXP-MAP-A01 — Mapping-Family Comparison

Priority: A

Evidence State: R/N

Associated Claims: MAP-001, MAP-002, MAP-003, MAP-004

Objective

Determine whether correspondence is specific to one arithmetic-to-geometric mapping or can be generated by many flexible mappings.

Candidate mappings

Historical candidates include:

$(\log|\Delta_E|,$
 $\log N_E,$
 $\alpha r_E),$
 $]$

and alternatives identified in:

[Criteria_for_Alternative_Mappings_Φ.pdf](#).

These include the MCJ, PTD, and FT families.

Comparison

Every mapping must be subjected to identical:

- data;
- null models;
- topological metrics;

- parameter-search budgets;
- validation splits.

Success Criterion

A mapping should demonstrate reproducible performance without requiring uncontrolled post hoc parameter tuning.

19. EXP-MAP-A02 — Rank-Scale Sensitivity

Priority: A

Evidence State: N

Historical mapping

```
[
z=200r.
]
```

This remains a historical phenomenological mapping:

```
[
h_r=\lambda_r r,
\quad
\lambda_r=200.
]
```

Objective

Determine whether the observed correspondence depends specifically on 200.

Test

Compare a pre-defined family:

```
[  
  \lambda_r\in\Lambda.  
]
```

The final range must be fixed before evaluating the test dataset.

Null

Random or independently generated scaling coefficients.

Success

Correspondence must remain significant across a meaningful range or exhibit a theoretically justified optimum that survives independent validation.

20. EXP-MAP-A03 — Arithmetic Generator Controls

Priority: A

Evidence State: N/R

Historical construction

Fibonacci and Lucas sequences have been used to construct arithmetic parameters.

Objective

Determine whether the observed correspondence depends on those specific sequences.

Controls

Compare:

- Fibonacci;
- Lucas;
- prime sequences;
- randomized recurrence sequences;

- matched-complexity alternative sequences.

Requirement

Sequence complexity and parameter count must be controlled as closely as possible.

Interpretation

A Fibonacci/Lucas result that is reproduced equally well by arbitrary matched-complexity sequences does not establish a special Fibonacci/Lucas correspondence.

21. EXP-DATA-A01 — Cross-Survey Matching Validation

Priority: A

Evidence State: U/R

Associated Claim: DATA-M001

Historical procedure

SDSS and MaNGA/HI data were joined using coordinate matching with a nominal tolerance of:

```
[  
  \theta_{\rm match}\leq 2".  
]
```

The field:

`arcsec_separation`

was retained as a diagnostic.

Objective

Quantify:

- match rate;
- false-match rate;

- separation distribution;
- dependence on source density;
- sensitivity to matching tolerance.

Controls

Compare:

```
[
1",\quad2",\quad3"
]
```

or another pre-registered tolerance family.

Success

Matching is reproducible and its uncertainty is characterized.

22. EXP-DATA-A02 — Data-Lineage Reconstruction

Priority: A

Evidence State: R

Objective

Construct the controlled chain:

```
[
D_{\rm raw}
\rightarrow
D_{\rm analysis}
\rightarrow
D_{\rm model}.
]
```

Required records

For every dataset:

- original source;
- acquisition date/version;
- file identifier;
- row count;
- column count;
- missingness;
- transformations;
- joins;
- matching tolerance;
- exclusions;
- derived columns;
- target definitions;
- code version.

Purpose

Ensure that every experimental result can be traced back to its source data.

23. EXP-RTCH-B01 — RTCH Mathematical Consistency

Priority: B

Evidence State: D

Objective

Audit the mathematical consistency of the proposed Relativistic Thermodynamic Cohomology framework.

Candidate ingredients

```
[  
F_Q=\Phi^*\omega,  
\quad  
dF_Q=0,  
]
```

with

[
[$\omega \neq 0$.
]

Candidate scalar:

$\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$.
]

Candidate action:

$S_{\rm EH}$
+
 $S_{\rm \phi}$
+
 $S_{\rm Y}$
+
 $S_{\rm B}$
+
 $S_{\rm top}$
+
 $S_{\rm m}$
+
 $S_{\rm GHY}$.
]

Required audit

Derive independently:

- metric variation;
- scalar equation;
- (B)-field equation;
- matter equation;
- conservation relation;
- point-particle limit;
- perfect-fluid limit.

Success

Equations follow consistently from the stated action and assumptions.

24. EXP-RTCH-B02 — RTCH Fifth-Force Constraint

Priority: B

Evidence State: N

Candidate prediction

With:

$A^2(\phi, \mathcal{M}, \mathcal{J}_Q)g_{\{\mu\nu\}},$
]

a point particle may experience:

$-c^2 h^{\{\mu\nu\}}$
 $\nabla_\nu \ln A.$
]

Objective

Determine whether the predicted deviation from geodesic motion is compatible with observational constraints.

Tests

Evaluate candidate effects in:

- Solar System;
- pulsars;
- galaxy dynamics;
- cosmology.

This experiment must be parameterized only after the mathematical consistency audit.

25. EXP-RTCH-B03 — Trace-Free Radiation Control

Priority: B

Evidence State: D/N

For a perfect fluid:

$$\frac{\epsilon+p}{c^2}u^\mu u^\nu + pg^{\mu\nu},$$

with:

$$[\bar{T} = -\epsilon + 3p.]$$

For radiation:

$$[p = \frac{\epsilon}{3},]$$

therefore:

$$[\bar{T} = 0.]$$

Objective

Determine the consequences of this result for the conformal coupling and associated fifth-force term.

Purpose

This provides a clean theoretical control case before applying the theory to matter-dominated systems.

26. EXP-FIELD-B01 — Arithmetic Field Construction

Priority: B

Evidence State: N

Candidate construction

$f(l_1(E), \dots, l_n(E)).$
]

Candidate gradient:

$-\nabla h_{\rm arith}.$
]

Objective

Determine whether an arithmetic field can be constructed mathematically without assuming in advance that it represents gravity.

Physical comparison

Compare against:

- matter density;
- gravitational potential;
- lensing;
- peculiar velocities;
- morphology;
- SFR;
- radiation.

Null

Random arithmetic fields matched in spatial statistics.

27. EXP-FIELD-B02 — Arithmetic Field vs Einstein Tensor

Priority: B

Evidence State: N

Objective

Determine whether an arithmetic-derived tensor has a mathematically and empirically meaningful relation to the Einstein tensor.

The experiment must not assume that the arithmetic field replaces:

[
 $G_{\mu\nu}$.
]

Instead test candidate relations:

[
 $G_{\mu\nu}$
 $\leftrightharpoons$
 $T_{\mu\nu}^{\text{arith}}$
]

under explicitly defined models.

Success

A relation must be:

1. mathematically well-defined;
 2. dimensionally consistent;
 3. independently fitted;
 4. independently tested;
 5. physically constrained.
-

28. EXP-ECC-B01 — Entropy/Cohomology Consistency

Priority: B

Evidence State: D/R

Candidate structure

Use a closed, non-exact two-form:

```
[  
d_{\mathcal T}\omega=0,  
\quad  
\omega\neq 0.  
]
```

Pull back:

```
[  
F_Q=\Phi^*\omega.  
]
```

Then:

```
[  
dF_Q=0.  
]
```

Objective

Verify the mathematical consistency of the entropy/cohomology construction and distinguish:

- exact forms;
- closed forms;
- closed non-exact forms;
- local scalar quantities;
- global cohomological quantities.

Important correction

The historical statement

[
 $d(\text{d}\mathcal{M})=0$
]

is not sufficient to establish nontrivial cohomology.

Nontrivial topology requires the appropriate cohomology class to be nonzero.

29. EXP-ENERGY-B01 — Generalized Mass-Energy Relation

Priority: B/C

Evidence State: N

Objective

Investigate whether the thermodynamic/cohomological framework can produce a generalized energy relation containing the known limit:

[
 $E=mc^2$.
]

Requirements

Any proposed relation must satisfy:

1. dimensional consistency;
2. known-limit recovery;
3. variational or mathematical origin;
4. conservation requirements;
5. stability;
6. well-posedness;
7. independent predictions.

Prohibition

A fitted equation reproducing existing data is not by itself a derivation of ($E=mc^2$) or a replacement for it.

30. EXP-COSMO-B01 — Arithmetic/Topological Evolution

Priority: B

Evidence State: N

Objective

Investigate whether arithmetic/topological structure evolves systematically with redshift.

Candidate comparison:

[
 $\mathcal{R}(\tau)$
 $\xrightarrow{\quad}$
 $D_{\text{cosmic}}(z)$.
]

Observables

- persistent Betti numbers;
- persistence landscapes;
- Wasserstein distances;
- rank distributions;
- environmental distributions.

Requirement

Evolutionary analysis must use independent redshift bins and avoid using future bins to construct earlier mappings.

31. EXP-COSMO-C01 — Hubble–Planck Tension Application

Priority: C/B after core validation

Evidence State: N

Objective

Determine whether a validated arithmetic/geometric/thermodynamic mechanism produces a quantitative modification relevant to cosmological expansion.

Prerequisite

This experiment cannot be used to justify the underlying framework.

The required sequence is:

```
[  
\text{mathematical construction}  
\rightarrow  
\text{independent evidence}  
\rightarrow  
\text{physical mechanism}  
\rightarrow  
H(z)  
\rightarrow  
\text{Hubble--Planck comparison}.  
]
```

Requirement

The model must make a quantitative prediction before comparison with the tension.

32. Statistical Standards

Every Priority A experiment must report, where applicable:

- effect size;
- uncertainty;
- confidence interval;
- empirical null distribution;
- (p)-value;

- multiple-comparison handling;
- predictive performance;
- calibration;
- sensitivity analysis;
- independent validation.

Statistical significance alone does not establish predictive usefulness.

The project therefore distinguishes:

[
 $\text{statistical correspondence}$
 $\neq$
 $\text{practical predictive power}$.
]

33. Machine-Learning Standards

All S.T.A.R. predictive experiments must distinguish:

Baseline

[
 $Y=f(X_{\text{astro}})$
]

Arithmetic-only

[
 $Y=f(X_{\text{arith}})$
]

Combined

[
 $Y=f(X_{\text{astro}}, X_{\text{arith}})$
]

Permutation

```
[
Y=f(X_{\rm astro},\pi(X_{\rm arith})).
]
```

Where appropriate, models should include:

- Random Forest;
- Gradient Boosting;
- other pre-registered baseline models;
- symbolic regression only as a discovery tool.

Symbolic regression output is a candidate relation, not a mathematical proof.

34. Cross-Validation and Independent Validation

Model development must be separated from final evaluation.

Where hyperparameter optimization is performed:

```
[
D_{\rm train}
\rightarrow
\text{CV/optimization}
\rightarrow
\text{locked model}
\rightarrow
D_{\rm test}.
]
```

Independent replication should use:

```
[
D_{\rm independent}
]
```

that was not used in:

- feature construction;
- mapping selection;
- hyperparameter selection;

- threshold selection;
 - model selection.
-

35. Null-Model Standards

Every core claim must have an explicit null.

Possible null families include:

N1 — Label permutation

Break correspondence while preserving marginals.

N2 — Spatial permutation

Preserve arithmetic distributions while destroying spatial association.

N3 — Matched random control

Generate a random control with matched sample size and distribution.

N4 — Alternative invariant

Replace rank with another arithmetic invariant.

N5 — Alternative mapping

Replace the proposed mapping with a controlled alternative.

N6 — Complexity-matched sequence

Replace Fibonacci/Lucas with sequences of comparable complexity.

N7 — Astrophysical baseline

Determine whether ordinary astrophysical variables already explain the effect.

36. Mapping Registry Integration

Every experiment involving (Φ) must reference a Mapping Registry entry.

At minimum record:

```
[  
  \Phi(E),  
]
```

the input invariants, normalization, free parameters, parameter-search range, and transformation.

The mapping must be fixed before final validation.

A mapping that is selected after observing the validation result is not considered independent validation.

37. Parameter Locking

Each experiment must distinguish:

Locked parameters

Parameters specified before final evaluation.

Training parameters

Parameters estimated from the training data.

Search parameters

Parameters explored during model selection.

Historical parameters

Parameters inherited from previous R&D but not yet validated.

Historical values such as:

```
[  
  \lambda_r=200  
]
```

must be explicitly marked historical until independently tested.

38. Reproducibility Requirements

A completed Priority A experiment must preserve:

1. source dataset identifier;
2. dataset checksum/version;
3. preprocessing code;
4. experiment code;
5. parameter file;
6. random seeds;
7. train/test split;
8. null-generation procedure;
9. output metrics;
10. plots/tables;
11. environment/package information;
12. experiment manifest.

A result that cannot be reconstructed from these artifacts remains historical evidence.

39. Results Classification

Experimental results should ultimately be classified as:

R0 — Failed/Unreproducible

Cannot reproduce the historical result.

R1 — Reproduced Historically

Historical result reproduced but not independently validated.

R2 — Controlled

Result survives leakage and null controls.

R3 — Replicated

Result survives an independent dataset.

R4 — Predictive

Result provides meaningful out-of-sample predictive performance.

R5 — Physical

Result is connected to a mathematically consistent physical theory and independently tested physical prediction.

These levels correspond to the broader S.T.A.R. success ladder.

40. Current Experimental Dependency Graph

The experiments should proceed approximately as:

```
[  
\boxed{\text{Data Provenance}}  
]
```

```
[  
\downarrow  
]
```

```
[  
\boxed{\text{EXP-DATA-A01/A02}}  
]
```

```
[  
\downarrow  
]
```

```
[  
\boxed{\text{EXP-CTRL-A03}}  
]
```

```
[  
\downarrow  
]
```

[
\\boxed{\\text{Astrophysical Baselines}}
]

[
\\downarrow
]

[
\\boxed{\\text{Arithmetic Feature Reconstruction}}
]

[
\\downarrow
]

[
\\boxed{\\text{EXP-CTRL-A01/A02}}
]

[
\\downarrow
]

[
\\boxed{\\text{EXP-SFR-A01/A02/A03}}
]

and in parallel:

[
\\boxed{\\text{EXP-K001}}
\\rightarrow
\\boxed{\\text{EXP-L001}}
]

and:

[
\\boxed{\\text{EXP-RANK-A01}}
\\rightarrow
\\boxed{\\text{EXP-RANK-A02}}
\\rightarrow
\\boxed{\\text{EXP-TOPO-A01}}.
]

Mapping validation runs across the topology and SFR programs:

[
\\boxed{\\text{EXP-MAP-A01/A02/A03}}.
]

Only after successful Priority A results:

[
\\boxed{\\text{Cross-Dataset Replication}}
]

[
\\downarrow
]

[
\\boxed{\\text{RTCH / FIELD / ECC / ENERGY}}
]

[
\\downarrow
]

[
\\boxed{\\text{Cosmological Evolution}}
]

[
\\downarrow
]

[
\\boxed{\\text{Hubble--Planck Application}}.
]

41. Historical R&D Integration

The existing R&D archive should not be discarded.

It becomes the historical experimental reservoir.

Historical scripts, datasets, figures, numerical results, failed models, alternative mappings, and exploratory analyses should be linked to experiment IDs.

The archive therefore answers:

“What has already been attempted?”

The Experiment Registry answers:

“What experiment do we now consider controlled?”

The Results Registry will answer:

“What did the controlled experiment actually show?”

The Replication Registry will answer:

“Did it survive independent validation?”

42. Immediate Experimental Queue

The initial controlled queue is:

Order	Experiment	Purpose
1	EXP-DATA-A01	Validate cross-survey matching
2	EXP-DATA-A02	Establish data lineage
3	EXP-CTRL-A03	Leakage audit
4	EXP-CTRL-A01	Random-feature baseline
5	EXP-K001	Reconstruct ($K_{\rm theory}$)
6	EXP-L001	Reconstruct ($L_{\rm cosmo}(s)$)
7	EXP-CTRL-A02	Arithmetic permutation control
8	EXP-SFR-A01	Incremental arithmetic SFR test
9	EXP-SFR-A02	Arithmetic-only SFR test
10	EXP-SFR-A03	Extreme-SFR robustness

11	EXP-RANK-A01	Rank/environment test
12	EXP-RANK-A02	Rank/topology test
13	EXP-TOPO-A01	Null-calibrated persistent homology
14	EXP-MAP-A01	Mapping-family comparison
15	EXP-MAP-A02	Rank-scale sensitivity
16	EXP-MAP-A03	Fibonacci/Lucas controls

Priority-B theoretical experiments begin only after the relevant mathematical and empirical dependencies have been resolved.

43. Registry Governance

No experiment may be marked **Complete** merely because a script ran successfully.

Completion requires:

```
[
\text{Specification}
+
\text{Data Provenance}
+
\text{Locked Parameters}
+
\text{Null Model}
+
\text{Execution}
+
\text{Result}
+
\text{Reproducibility}.
]
```

No historical result may be silently upgraded to evidence.

No statistically significant result may be silently upgraded to predictive usefulness.

No predictive result may be silently upgraded to physical theory.

No physical interpretation may be treated as established without independent validation.

44. Version 0.2 Status

Experiment Registry v0.2 establishes the controlled experimental architecture.

It incorporates the historical R&D reservoir while explicitly separating:

[
 $\boxed{\text{Historical R\&D}}$
]

from

[
 $\boxed{\text{Reconstructed Evidence}}$
]

and

[
 $\boxed{\text{New Controlled Experiments}}$.
]

The next registry required is the **Data Provenance Registry**, constructed from the datasets actually referenced by the Experiment Registry.

The intended operational chain is therefore:

[
 $\boxed{\text{Claim} \rightarrow \text{Evidence} \rightarrow \text{Experiment} \rightarrow \text{Data} \rightarrow \text{Result} \rightarrow \text{Replication}}$
]

}
]

This chain is the primary mechanism for preserving the scientific integrity and continuity of the S.T.A.R. research program.